

汪露

AI软件工程师

13122038365 (微信同号) | wanglu_ashley@126.com

专注于生产级 LLM 应用与 AI Agent 系统建设。擅长基于 Python/Java 构建高可用后端服务，在 Agent Workflow、Tool Calling、Streaming、个性化推荐、商品决策辅助及 Text2SQL 等方向具备丰富实践经验。主导电商与金融领域 AI 产品研发，负责个性化导购、实时响应优化、企业级数据分析 Agent 等核心能力建设，具备从系统设计到生产落地的完整经验。

工作经历

2025.08-至今	AI软件工程师	嘉兰朵（深圳）科技发展有限公司
2021.04-2025.08	AI软件工程师	思特沃克软件技术（深圳）有限公司

嘉兰朵/Zalando项目经验

个性化推荐与导购能力建设

面向 Zalando Assistant 的个性化推荐与导购场景，负责首屏 Conversation Starters 和商品详情页导购推荐能力建设。结合用户画像、历史行为、对话上下文和商品信息生成个性化推荐内容。核心挑战在于兼顾推荐效果、业务可控性与高并发场景下的性能和稳定性。

核心职责：

- **个性化输入链路构建：** 构建“历史行为 + 对话上下文 + 用户画像”的个性化输入链路，解决原推荐逻辑仅依赖静态行为数据、缺乏对话连续性感知的问题，提升推荐相关性与新特性曝光率，推动推荐入口互动率提升**15%+**。
- **画像服务缓存层设计：** 在不改动原有业务调用链路的前提下，为下游用户画像服务增加缓存层，实现 TTL 缓存、字段级失效、异常隔离和空值过滤能力，使下游用户画像服务调用量减少**70%+**，高并发场景下接口 P99 延迟下降约**60%**。
- **Warm-Up架构设计与落地：** 设计并落地基于**Redis 注册表 + 内存回退机制**的异步预计算架构，将首屏对话引导建议生成任务与用户主请求解耦，使耗时建议生成可在后台提前完成，首屏冷启动时间降低 **60%+**。
- **推荐策略框架设计与落地：** 主导商品详情页个性化导购推荐策略重构，设计“行为信号识别 + 推荐方向决策 + 兜底生成”的多层决策框架，解决纯生成不可控、静态推荐模板个性化不足等问题，保障不同用户与商品场景下均可稳定产出高相关推荐问题。
- **真实场景驱动的策略优化：** 基于约800个真实商品详情页场景建立推荐效果分析体系，识别覆盖不足、角度重复和低质量入口等问题，并将分析结果转化为推荐规则与安全门控策略，推动推荐策略从经验驱动向数据驱动演进。

Agent Runtime 主链路升级与实时响应优化

负责商品详情 Tool 接入、Streaming 架构优化及 OpenAI Responses API 迁移。核心挑战在于多工具调用、业务事件和模型输出在同一链路中的协同处理，以及实时响应性能与功能一致性的保障。

核心职责：

- **商品详情 Tool 接入与 Agent 编排：**设计并实现商品详情 Tool 接入方案，使 Agent 能够根据用户意图动态触发工具调用并整合实时商品信息生成导购回答，解决纯模型回答缺乏事实支撑的问题。
- **Streaming 响应架构设计与延迟优化：**设计并实现复杂多步 Agent 链路下的 Streaming 处理与分发机制，基于状态机区分过程状态、业务事件和最终回复文本，解决多工具调用场景下的重复展示与时序错乱问题。基准测试中，Suggestions API 流式输出相较原同步接口，TTFT 平均延迟降低约**25%**。
- **OpenAI Responses API 迁移与扩展能力建设：**完成 Agent 主链路从 Chat Completions 向 Responses API 的迁移，统一 Tool Calling 与 Streaming 处理链路，为后续状态复用、多工具协作和新模型接入提供扩展能力。在已验证链路中，P95 TTFT降低约**25%**。

Product Comparison Skill 商品对比能力

设计并上线 Product Comparison Skill，帮助用户通过自然语言完成多商品比较与决策。核心挑战在于多轮对话中的商品引用解析，以及兼顾模型理解能力与前端稳定渲染要求。

核心职责：

- **商品对比Skill设计与上线：**设计并上线 Product Comparison Skill，将开放式自然语言对比请求抽象为可复用能力，支持不同导购场景下的商品比较与决策辅助，上线后对比场景用户互动率提升**20%+**。
- **多轮商品引用解析：**构建上下文商品解析和对比对象维护机制，解决用户通过序号、范围或追加表达引用商品时的对象定位问题，并在对比对象变化后完整重建结果，避免长对话中的状态混乱。
- **动态对比维度决策：**结合用户关注点、商品属性和品类特征动态选择比较维度，支持价格、品牌、材质、舒适度等差异化信息展示，使不同场景下的比较结果更加聚焦和可读。
- **结构化输出稳定性治理：**拆分模型推理与表格生成职责，由模型负责理解意图和生成总结建议，由代码负责字段解析、折扣计算、两商品/多商品布局和渲染标记，解决纯 LLM 生成表格格式不稳定、字段缺失和前端难以渲染的问题。

思特沃克/Thoughtworks项目经验

定价管理系统AI Agent模块 国内知名银行

该项目基于 AIGC 技术，服务于银行内部员工，目标是提升非技术人员的数据分析与政策查询能力。核心功能包括：通过 **Text2SQL** 实现结构化数据的自然语言查询，以及基于知识库的政策类问答。作为核心开发工程师，负责数据分析模块的整体设计与落地，包括数据分析链路构建、SQL 生成优化与系统联调上线。

核心职责：

- **Agent 应用开发：**主导智能 Agent 应用开发，设计并实现“意图澄清 → SQL生成 → 查询执行 → 结果总结与可视化”的多阶段 Agent 工作流，支持复杂查询、多轮澄清和分析结果自动解读。
- **Prompt 工程与效果评估：**针对复杂业务查询场景持续迭代 Prompt 模板与推理流程，构建 1000+ 条真实业务测试集用于效果评估和回归验证，提升模型端到端问答准确率 **20%+**，增强模型对复杂查询意图的理解能力和 SQL 生成稳定性。
- **SQL 生成优化：**梳理高频业务语料，并基于向量数据库构建字段+值双层召回的 Rerank 机制；引入 SQL 校验、异常重试等机制，增强整体链路稳定性。

RAG智能问答Chatbot - 核心开发工程师

基于 LlamaIndex + FAISS 构建文档切片、向量化、召回与答案生成链路，并引入 RAGAS 评估答案相关性和上下文覆盖率，形成可复用的 RAG 研发与评估模板。

定价管理系统（权限与报表） – 模块负责人

主导基于六边形架构+DDD的权限模块设计，自研内存数据加工方案（Tablesaw），报表开发效率提升**约40%**；推动MySQL到新架构的数据库迁移，实现0业务感知、每日数百万笔定价请求稳定运行。

售后管理系统 – 模块负责人 & 核心后端工程师

负责售后模块后端架构与API对接，协调总部、中台、WMS多系统集成；所负责模块准时交付，联调阻塞修复率约98%，系统上线后支撑日均数万级备件交易流转。

技能

- **AI / LLM 应用开发：**LLM Agent、Agent Workflow、Pydantic-AI、Tool Calling、OpenAI Responses API、Streaming、RAG、Text2SQL、Prompt Engineering、RAGAS
- **检索与向量技术：**FAISS、LlamaIndex、Embedding、Rerank（BGE-Reranker）、pgvector
- **编程&后端：**Python、Java、FastAPI、Spring Cloud、Redis
- **工程化工具：**Docker、Git、Pytest

教育背景

- 英国卡迪夫大学 Computing 2019-2021 硕士
- 上海工程技术大学 市场营销 2015-2019 本科

Lu Wang

wanglu_ashley@126.com

AI Software Engineer

AI Software Engineer focused on building LLM applications and AI Agent systems for real business scenarios. Experienced in Python/Java backend development, Agent Workflows, Tool Calling, Streaming, personalization, product decision support, Text2SQL, and observability-driven performance diagnostics. Led AI product capabilities across e-commerce and financial services, including personalized shopping assistance, real-time response optimization, and enterprise data analysis agents, with end-to-end experience from system design to production rollout.

Work Experience

2025.08 - Present

AI Software Engineer

Zalando Technology Development (Shenzhen) Co., Ltd.

2021.04 - 2025.08

AI Software Engineer

Thoughtworks Software Technologies (Shenzhen) Co., Ltd.

Zalando Experience

Personalization and Guided Shopping Capabilities

Worked on personalization and guided-shopping capabilities for Zalando Assistant Conversation Starters. The system combines user profiles, behavioral history, conversation context, and product information to generate personalized recommendation content. The main challenge was improving recommendation relevance while maintaining compliance with business rules and ensuring performance and stability under high traffic.

Key Contributions

- **Built a personalization input pipeline** combining user behavior, conversation context, and profile signals, addressing the previous reliance on static behavioral data and increasing Conversation Starter engagement by **15%+**.
- **Designed a cache layer** for downstream user profile data with TTL control, field-level invalidation, failure-safe handling, and null-value filtering, reducing profile-service calls by **70%+** and lowering P99 latency by about **60%** under high concurrency.
- **Implemented an asynchronous Warm-Up architecture** using Redis registry and in-memory fallback, decoupling first-screen Conversation Starter generation from user requests and reducing cold-start time by **60%+**.
- **Redesigned the product-detail-page recommendation strategy** with a multi-layer decision flow covering behavior-signal recognition, recommendation routing, and fallback generation, improving recommendation stability across different user and product scenarios.
- **Built an evaluation process** based on about **800** real product-detail-page scenarios, identified coverage gaps and repeated recommendation angles, and translated findings into recommendation rules and safety controls.

Agent Runtime Upgrade and Real-Time Response Optimization

Upgraded the main Zalando Assistant Agent Runtime by integrating product-detail tools, improving Streaming response handling, and migrating to OpenAI Responses API. The work focused on coordinating tool results, intermediate states, and model output in one real-time response flow.

Key Contributions

- **Integrated product-detail Tool Calling** into the Agent Runtime, enabling the Agent to retrieve real-time product information based on user intent and generate more grounded shopping responses.
- **Designed Streaming processing** for multi-step Agent flows, separating process states, business events, and final response text to prevent duplicated display and event-ordering issues in multi-tool scenarios.
- **Reduced Suggestions API** average TTFT by about **25%** in benchmark tests compared with the previous synchronous interface.
- **Migrated the main Agent Runtime** from Chat Completions to OpenAI Responses API, unifying Tool Calling and Streaming handling and reducing P95 TTFT by about **25%** in validated flows.

Product Comparison Skill

Designed and launched Product Comparison Skill for Zalando Assistant, enabling users to compare multiple products through natural language and make purchase decisions more efficiently. The core challenge was resolving product references across multi-turn conversations while producing structured comparison output that could be reliably rendered by the frontend.

Key Contributions

- **Designed and launched Product Comparison Skill**, turning open-ended natural-language comparison requests into a reusable shopping decision capability and increasing comparison-scenario engagement by **20%+**.
- **Built context-aware product reference** parsing to resolve product selections across multi-turn conversations, including position-based, range-based, and follow-up comparison requests.
- **Developed dynamic comparison-dimension selection** based on user intent, product attributes, and category characteristics, making comparison results more focused and easier to scan.
- **Separated model reasoning from table generation**, using the model for intent understanding and summary recommendations while deterministic code handled field parsing, discount calculation, layout, and rendering markers.

Thoughtworks Experience

AI Agent Module for Pricing Management System

To empower internal business users with self-service data access, the bank initiated an GenAI project supporting natural language queries for complex data analytics and internal policy Q&A. As a core engineer for the data analysis module, I was responsible for architecting the agent-based workflow, optimizing query generation accuracy, and delivering the solution to production.

Key Contributions

- **Designed and implemented** an intelligent agent workflow with three stages: intent clarification, Text2SQL generation, and result summarization with visualizations.
- **Created and optimized** prompt templates for complex business queries, improving model accuracy by **~20%** and enhancing its ability to handle nested conditions and multi-field constraints.
- **Developed a test suite with 1,000+** real-world business cases, engineered a dual-layer reranking mechanism (field + value) using vector search, and integrated SQL validation, exception handling, and automatic retry logic to enhance robustness.

RAG Enterprise Chatbot – Knowledge Q&A System

Core Developer

Built a reusable RAG pipeline with LlamaIndex and FAISS, covering document chunking, vectorization, retrieval, and answer generation; introduced RAGAS to evaluate answer relevance and context coverage.

Pricing Management System: Access Control and Reporting

Module Owner

Led access-control module design based on Hexagonal Architecture and DDD, built a Tablesaw-based in-memory reporting solution, improved reporting development efficiency by about **40%**, and supported a zero-impact database migration for millions of daily pricing requests.

After-Sales Management System

Module Owner & Core Backend Engineer

Owned backend architecture and API integration for the after-sales module across headquarters systems, platform services, and WMS; delivered the module 100% on time, resolved 98% of integration blockers, and supported tens of thousands of daily spare-part transactions after launch.

Skills & Technologies

- **AI / LLM Application Development:** LLM Agent, Agent Workflow, Pydantic-AI, Tool Calling, OpenAI Responses API, Streaming, RAG, Text2SQL, Prompt Engineering, RAGAS
- **Retrieval and Vector Technologies:** FAISS, LlamaIndex, Embedding, Reranking, BGE-Reranker, pgvector
- **Programming and Backend:** Python, Java, FastAPI, Spring Cloud, Redis
- **Engineering Tools:** Docker, Git, Pytest

Education

MSc in Computing, **Cardiff University**, UK
2019 – 2021

BA in Marketing, **Shanghai University of Engineering Science**, China
2015 – 2019